# LAB 4 — Chat History

## Vibe Coding / Lunar Lander Project

This PDF is a reconstructed transcript from the conversation content available in the current ChatGPT session. It preserves the project workflow, commands, code changes, testing, commits, and GitHub steps that are available in the conversation context. It does not claim to reproduce any messages or attachments that are not available in the current session.

**USER**

I need to clone the assigned GitHub repository https://github.com/SETAPESU26/32_lunarLander to my own GitHub, fix errors and run the game. I have 0 knowledge and want to do it from scratch on VS Code.

**ASSISTANT**

We started from VS Code and the user's GitHub account. The repository was cloned into D:\SEM 5\SE\vibecoding\32_lunarLander.

**USER**

The VS Code terminal showed the repository cloning successfully, followed by Python 3.14.7 and pip 26.2.1.

**ASSISTANT**

We checked the Python and pip installation, then installed pygame-ce using pip. The installation succeeded and pygame-ce 2.5.8 was verified with Python.

**USER**

The game opened and the before video was recorded.

**ASSISTANT**

We began Task 1. The README requirement was to fix landing validation by checking the horizontal speed limit as well as vertical speed and angle.

**USER**

The original touchdown() function was provided. It checked pad presence, vertical speed and angle, but did not check horizontal speed.

**ASSISTANT**

The missing condition was identified. The landing condition was changed from:

if pad and abs(self.vel.y) <= MAX_SPEED_Y and abs(angle) <= MAX_ANGLE:

to:

if pad and abs(self.vel.x) <= MAX_SPEED_X and abs(self.vel.y) <= MAX_SPEED_Y and abs(angle) <= MAX_ANGLE:

**USER**

The git diff showed the horizontal-speed check being added. Then git add game.py and git commit -m "Fix landing horizontal speed validation" were run successfully.

**ASSISTANT**

Task 1 was confirmed as committed successfully.

**USER**

README Task 2: Implement ship_color(fuel_ratio). It is called once per frame in draw, receives fuel from 0.0 to 1.0, and should return an RGB hull color or None. The idea is to shift the hull toward red as fuel runs low.

**ASSISTANT**

We implemented a smooth fuel-based color transition. The function clamps fuel_ratio between 0 and 1, keeps red at 230, and changes green and blue according to fuel level:

```
fuel_ratio = max(0.0, min(1.0, fuel_ratio))
r = 230
g = int(60 + 170 * fuel_ratio)
b = int(60 + 180 * fuel_ratio)
return (r, g, b)
```

**USER**

The git diff showed the pass being replaced by the fuel-color implementation.

**ASSISTANT**

After exiting the git diff viewer with q, Task 2 was committed with:

```
git add game.py
git commit -m "Add fuel based ship color"
```

The commit succeeded.

**USER**

README Task 3: Implement on_landing(score). It is called after a successful landing and can add fireworks or a distinct sound for a high-multiplier pad.

**USER**

The original on_landing(score) function was provided:

```
def on_landing(score):
"""Called after a successful landing with the points just earned; add fireworks or bonuses
here."""
pass
```

**USER**

The draw(self, screen) function was provided so the landing effect could be integrated with the existing drawing loop.

**ASSISTANT**

A simple fireworks particle effect was implemented. A global landing_particles list was added. on_landing(score) creates 35 particles with random positions, directions, speeds and colors. draw() updates and renders the particles as circles and removes them after their lifetime expires.

**ASSISTANT**

The user tested the landing effect and later confirmed Task 3 was committed successfully.

**USER**

README Task 4: Implement bonus_life_threshold(). It is called every frame in update(), takes no arguments, returns an integer score threshold or None, and the existing self.bonus_awarded bookkeeping handles awarding a life. The README suggests returning 1500.

**ASSISTANT**

The required implementation is:

```
def bonus_life_threshold():
return 1500
```

This makes a bonus life occur when the score crosses 1500-point multiples, using the existing bookkeeping.

**USER**

During testing, a landing at low apparent speed crashed.

**USER**

A screenshot showed Fuel 42, Vx -26.0 in red, Vy 34.0 in green, Angle 0, and the message 'Crashed: too fast!'.

**ASSISTANT**

The screenshot was diagnosed: abs(Vx) = 26, while MAX_SPEED_X = 25. Therefore the new Task 1 horizontal-speed condition correctly rejected the landing. No code change was needed. The user was advised to reduce horizontal velocity below 25 before touching the pad.

**USER**

The user successfully landed after adjusting the velocity.

**ASSISTANT**

Task 1 was therefore confirmed to work correctly, and the user proceeded with Task 4 and GitHub setup.

**USER**

The personal GitHub repository was created.

**ASSISTANT**

The local Git remote was changed from the assigned SETAPESU26 repository to the user's repository:

git remote set-url origin https://github.com/Ayush-Gaikwad/32_lunarLander.git

Then git remote -v was used to verify the new origin.

**ASSISTANT**

The four task commits were checked with git log --oneline --max-count=4, then the project was pushed with:

git push -u origin main

**USER**

The terminal confirmed the push succeeded:

To https://github.com/Ayush-Gaikwad/32_lunarLander.git
* [new branch] main -> main

branch 'main' set up to track 'origin/main'.

**USER**

The after video was also recorded.

**ASSISTANT**

At this point the coding work and before/after videos were complete. The remaining submission requirement discussed was the chat-history PDF and organizing the Lab-4 submission folder.

**USER**

The user reported that browser Print → Save as PDF was missing some parts of the conversation.

**ASSISTANT**

The recommended solution was to create a clean PDF from the conversation content rather than relying on the browser's print rendering. This PDF is that reconstructed version, based only on the conversation content available in the current session.